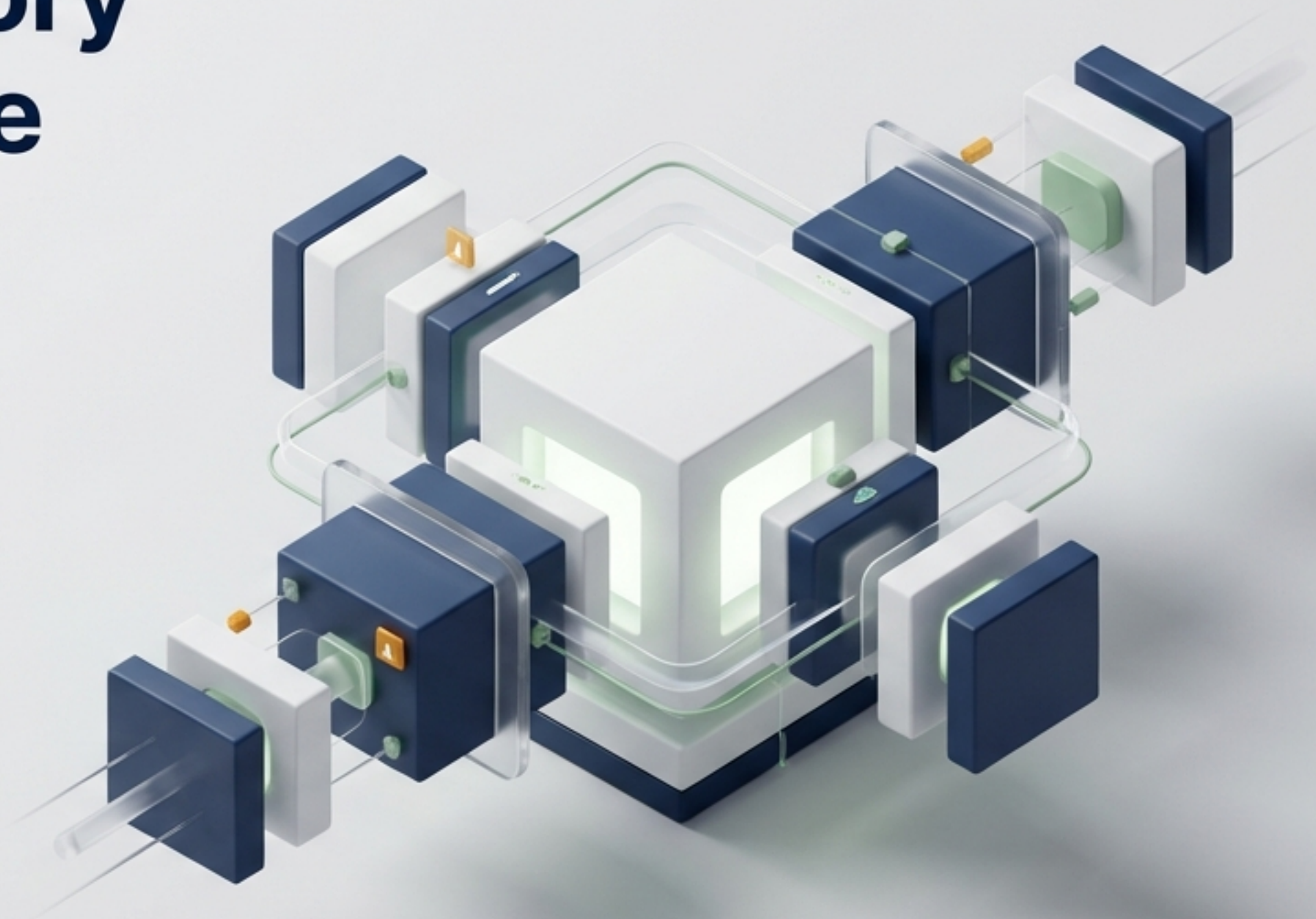


Advancing Regulatory Intelligence with the askFDALabel-Suite

A unified AI-augmented platform designed to accelerate clinical safety reviews, uncover pharmacovigilance signals, and harmonize drug labeling.



DRUG LABELING DENSITY HAS OUTPACED TRADITIONAL RETRIEVAL TOOLS

Drug labeling is the authoritative source of truth for safe and effective use. However, the complexity of this data has grown exponentially.

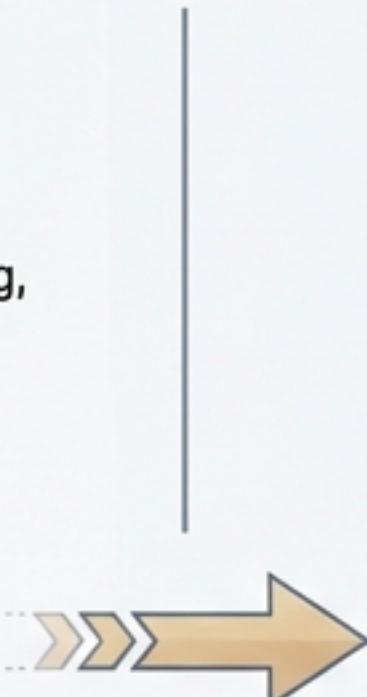
While traditional platforms like DailyMed and FDALabel remain the gold standard for retrieving individual labels, they are designed for retrieval, not high-volume synthesis.

Reviewers face a structural bottleneck when attempting multi-label reasoning, comparative analysis, or tracking evolving safety profiles across entire therapeutic classes.

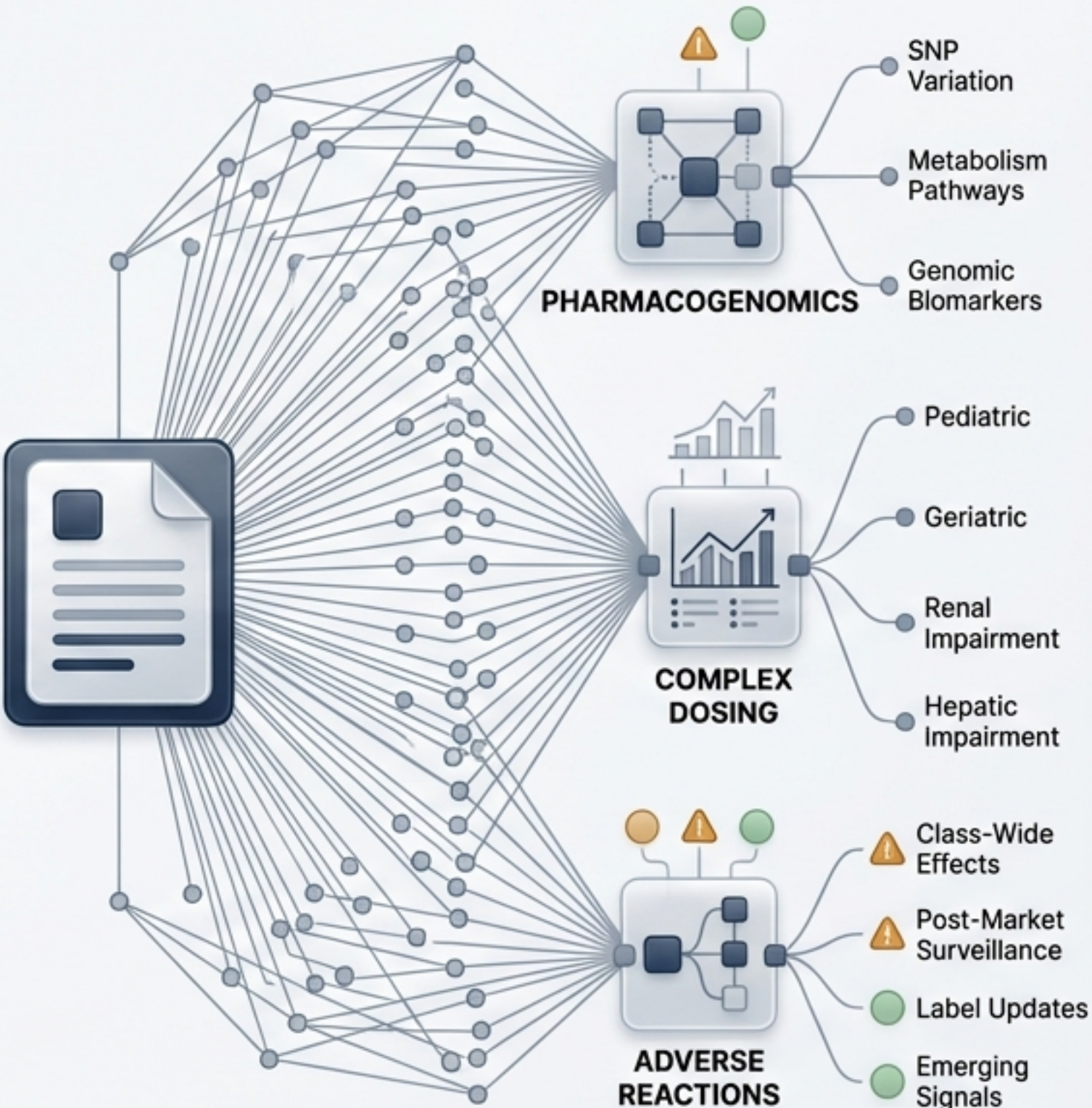


TRADITIONAL RETRIEVAL TOOLS

DailyMed & FDALabel | Single Label Focus



MODERN DATA SYNTHESIS & REASONING



MODERN DATA SYNTHESIS & REASONING

Multi-Label Analysis | Comparative Reasoning | Evolving Safety Profiles

Manual synthesis creates critical vulnerabilities in safety oversight

Traditional “search-and-read” workflows force clinical reviewers into time-intensive manual labor, leading to three systemic risks:

The Safety-Labeling Gap



Temporal lags between real-world safety signals emerging in the population and formal inclusion in product labeling.

Generic Alignment Lag



Discrepancies where generic manufacturers fail to synchronize updated toxicity warnings with the Reference Listed Drug (RLD).

Cognitive Fatigue

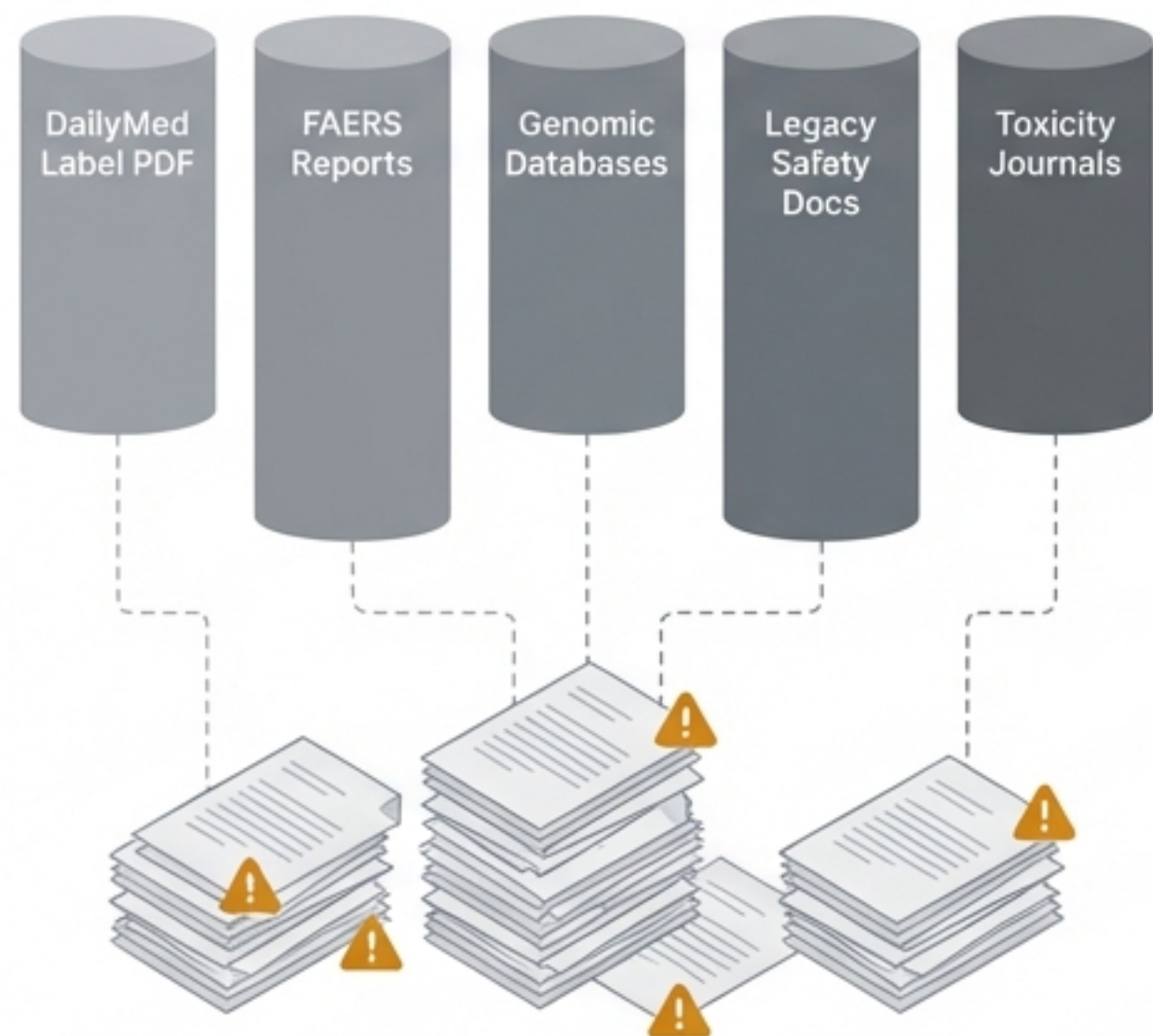


High susceptibility to human error when extracting complex genomic or dosing data across hundreds of dense XML documents.

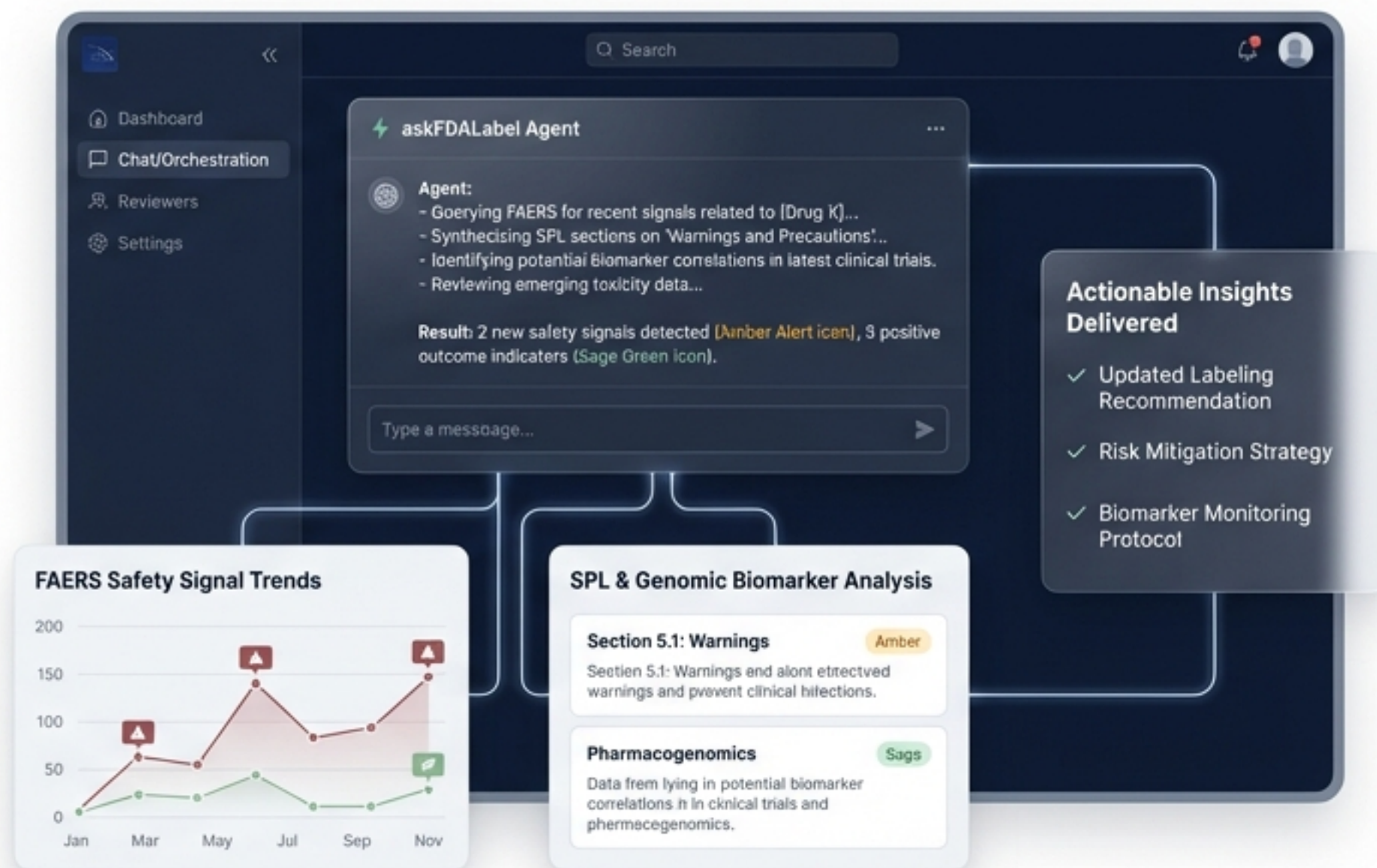
A paradigm shift from manual retrieval to AI-augmented synthesis

The askFDALabel-Suite bridges high-fidelity Structured Product Labeling (SPL) data with state-of-the-art LLM orchestration. It consolidates disparate regulatory tasks—safety signal detection, toxicity monitoring, genomic biomarker analysis, and structural label comparison—into a single environment. Reviewers are equipped with a powerful agentic assistant capable of performing complex clinical reasoning and delivering actionable, evidence-based insights.

Search-and-Read

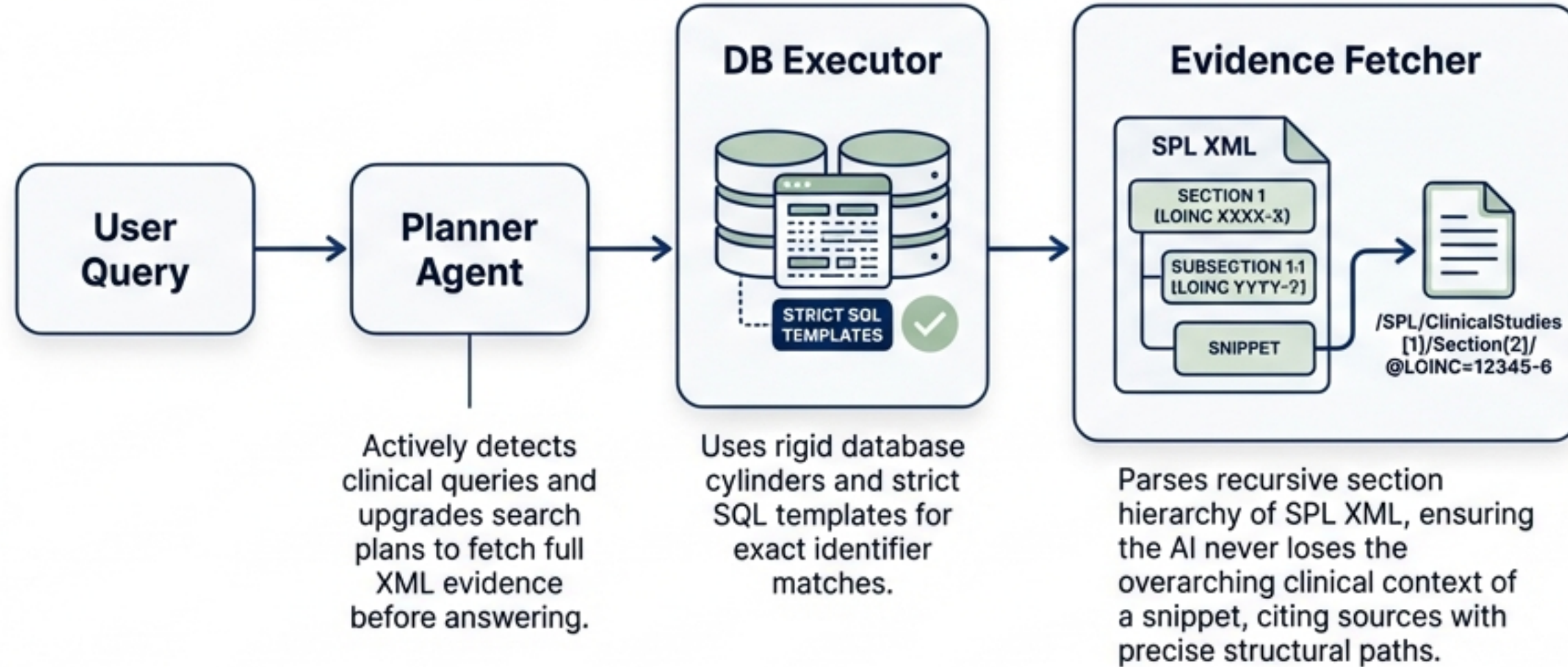


Query-and-Synthesize



Multi-step agentic reasoning grounded in structural evidence

To ensure regulatory-grade accuracy, the AFL Agent v2 avoids generic semantic search and relies on exact identifier matches and LOINC-to-Section mapping.

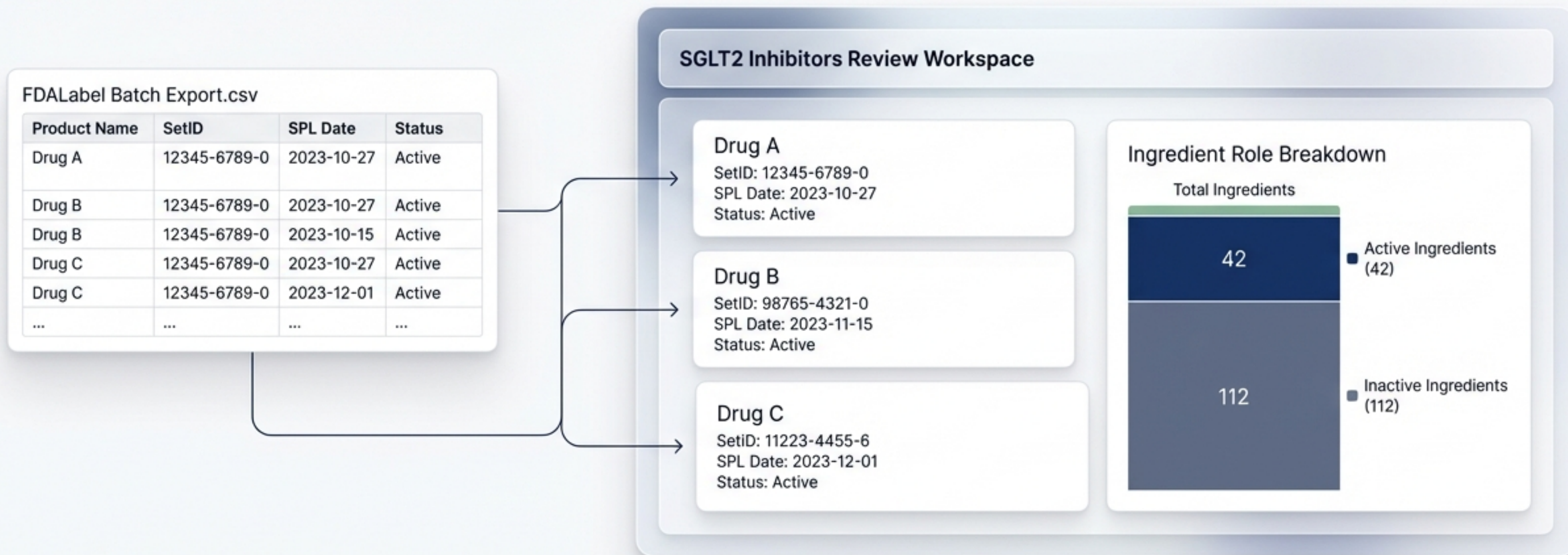


-  **Plan Upgrading:** The agent actively detects clinical queries and upgrades search plans to fetch full XML evidence before answering.
-  **Hierarchical Integrity:** By parsing the recursive section hierarchy of SPL XML, the system ensures the AI never loses the overarching clinical context of a snippet.
-  **Evidence-Based Execution:** The agent cites its sources with precise structural paths, eliminating context drift and mitigating hallucination risks.

Curated project workspaces for aggregate clinical analysis

The suite complements FDALabel's gold-standard search by allowing reviewers to batch-import complex queries directly into persistent project containers.

- **Automated Hydration:** Instantly fetches corresponding SPL XML content and hydrates the project with full clinical metadata upon import.
- **Duplicate Cleanup:** Internal logic automatically detects and removes duplicate SetIDs, ensuring subsequent analyses use a clean product census.
- **Ingredient Role Breakdown:** Instantly generates a quantitative census distinguishing Active vs. Inactive ingredients across the entire project for excipient safety auditing.



Correlating FAERS reporting with labeling to close safety gaps

The AE Profile module bridges official manufacturer labeling with real-world clinical reporting from openFDA.

A two-phase asynchronous scan isolates high-priority safety signals:

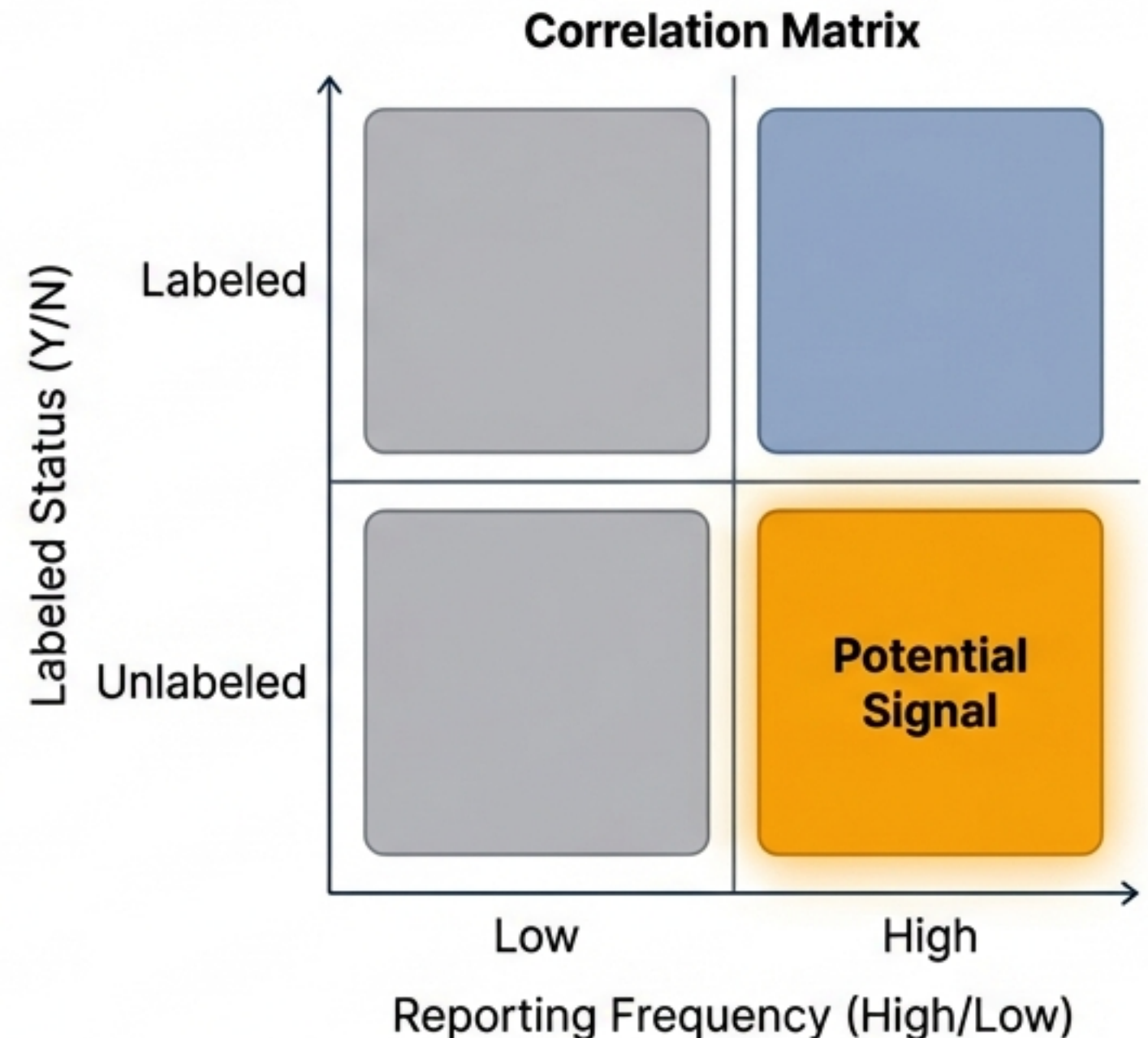


1. **Deep Labeling Scan:** Semantic, MedDRA-mapped extraction across high-impact safety sections to establish "Labeled Status."



2. **FAERS Integration:** Parallel retrieval of report counts across 1-year, 5-year, and all-time horizons to establish "Reported Status."

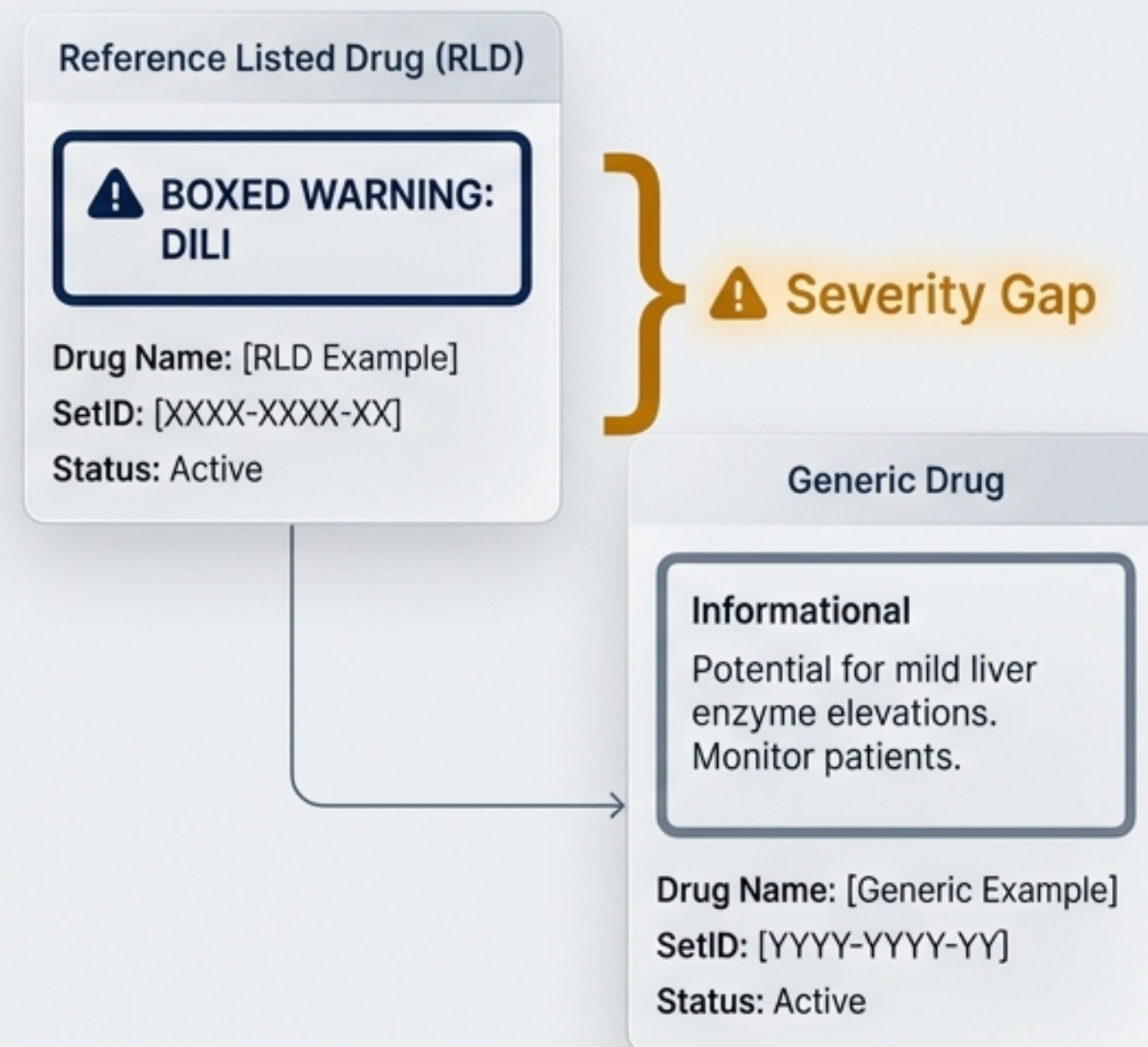
Outcome: Immediately flags under-labeled risks and class-wide signals for regulatory action.



Monitoring organ toxicity and generic severity gaps

The askDrugTox module is a specialized toxicology intelligence agent focused on Drug-Induced Liver Injury (DILI) and critical endpoints.

- **Severity Ranking:** Automatically parses language in Warnings and Precautions to rank toxicity severity (from Informational to Boxed Warning).
- **RLD Alignment:** Flags “Severity Gaps” where a generic manufacturer’s label lags behind the Reference Listed Drug.
- **Biomarker Correlation:** Maps labeling text to known toxicological biomarkers (e.g., ALT, AST elevations) to define the clinical thresholds triggering warnings.



Standardizing and extracting genomic biomarkers

As medicine moves toward a personalized model, the PGx Agent ensures critical genomic markers are never overlooked during a review.

- **Variant Extraction:** Deep parsing identifies specific alleles and links them to required dosing adjustments or contraindications.
- **Regulatory Categorization:** Automatically classifies the regulatory strength of the biomarker as Genetic Testing Required, Recommended, or Informational.
- **Clinical Synthesis:** Generates concise summaries explaining the "Clinical Why" (e.g., increased toxicity risks for poor metabolizers).

...patients with altered metabolism may experience reduced efficacy or increased toxicity. Testing for specific variants in **CYP2D6** is essential for determining appropriate dosing regimens for select therapeutics, as poor metabolizers are at higher risk. **CYP2D6** Additionally, screening for **HLA-B*1502** is crucial prior to initiating treatment with carbamazepine due to the risk of severe cutaneous adverse reactions (SCARs)...



Genetic Testing Required ⚠️

Mandatory for safety (e.g., avoid SCARs).



Recommended ✓

Improves efficacy or safety outcomes.



Informational ⓘ

Provides additional clinical context.

Detecting subtle safety drift across complex complex document structures

LabelComp is a high-precision comparison engine designed for Brand vs. Generic audits and tracking historical labeling evolution.

-  **Structural PLR Diffing:** Uses LOINC codes to ensure Section 5.1 in Label A is strictly compared against Section 5.1 in Label B.
-  **Nuanced Word-Level Diffing:** Optimized for clinical prose, ignoring formatting changes while highlighting alterations in warning severity.
-  **AI-Synthesized Summaries:** Bypasses trivial text changes to provide an executive summary of the actual clinical meaning and actionable changes between 50-page labels.

Older Label Text (Reference 2019)	Updated Label Text (Current 2023)
5.1 WARNINGS AND PRECAUTIONS: Pediatric Use: Safety and effectiveness in pediatric patients have not been established. Monitor for adverse reactions. Dosing adjustments may be required in renal impairment.	5.1 WARNINGS AND PRECAUTIONS: Pediatric Use: Safety and effectiveness in pediatric patients 12 and older have been established. Monitor for new adverse reactions, including Stevens-Johnson syndrome. Dosing adjustments are required in moderate to severe renal impairment.

 **Clinical Bottom Line:**
Revised pediatric indications and new adverse reactions

Meeting reviewers seamlessly within their active workflows

The suite utilizes Elsa Addons to bring deep retrieval capabilities directly into the FDA's internal AI chat ecosystem, eliminating cognitive fatigue from context switching.

- **Seamless Data Bridging:** Bookmarklets automatically parse the DOM on DailyMed/FDA Label to identify SetIDs, NDCs, or Application Numbers.
- **Instant Handshake:** Automatically primes the Elsa environment with the relevant labeling XML and clinical metadata.
- **Active Continuity:** Reviewers can launch complex labeling queries directly into an active chat thread without manual copy-pasting.



Injecting verifiable clinical trust into general AI interfaces

Elsa Addons provide semantic enrichment to ensure AI responses meet the exact standards of regulatory science.

- ✨ • **Dynamic Term Tagging:** Automatically highlights clinical entities like ingredients, biomarkers, and adverse reactions in real-time.
- 🔗 • **Intelligent Hover Cards:** Provides instant clinical context (e.g., project-wide role breakdown or regulatory testing status) without leaving the chat.
- 🔗 • **Embedded Evidence Links:** Creates an immutable **audit trail** by linking every highlighted term directly back to its specific section in the source SPL hierarchy.

Based on the SPL, the

Active Ingredient ↔ concentration is 10 mg/mL, consistent with the reference label.

Active Ingredient



Source: SPL Section 11.1 (XML Path: /document/component/structuredBody/component/section[code/@code='123-4']/text/paragraph[2])

Accelerating reviews and ensuring regulatory integrity

The askFDALabel-Suite replaces manual data extraction with decisive, **AI-augmented** regulatory action.



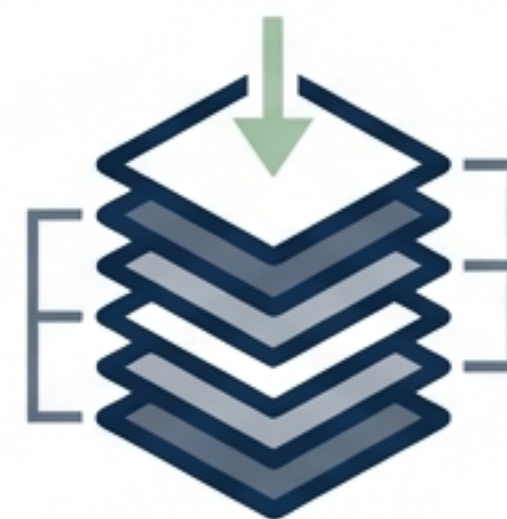
Closed Safety Gaps:
Instantly flag discrepancies between real-world FAERS reporting and established labeling.



Harmonized Therapeutics:
Ensure class-wide consistency and automatically detect severity gaps between generics and RLDs.



Precision PGx Tracking:
Standardize and extract complex genomic biomarkers to inform precision prescribing guidelines.



Faster Aggregate Analysis:
Transform chaotic, unstructured XML into clean, quantifiable project censuses for rapid class reviews.

Empowering reviewers to focus on science, not searching

By fundamentally shifting the workflow from manual retrieval to AI-augmented synthesis, the askFDALabel-Suite functions as an invisible, evidence-based partner.

It drastically reduces cognitive fatigue, guarantees structural exactness, and ensures that the FDA's top experts spend their time making the critical regulatory decisions that protect public health.

